PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Worldwide Prevalence, Risk Factors, and Psychological Impact of Infertility among Women: A Systematic Review and Meta-Analysis
AUTHORS	Nik Hazlina, Nik Hussain; Norhayati, Mohd Noor; Shaiful Bahari, Ismail; Nik Ahmad, Nik Muhammad Arif

VERSION 1 – REVIEW

REVIEWER	Howard Cabral Boston University School of Public Health, Department of Biostatistics
REVIEW RETURNED	22-Sep-2021

GENERAL COMMENTS	This manuscript presents the results of a random effects model-based meta-analysis of international studies in the literature that examined both psychological distress and infertility. The methodology for the analysis of the data is for the most part consistent with accepted practice, though the I2 value to justify the use of the random effects model was not included in the text and the funnel plots mentioned in the Methods were not shown. The generalization of the results based on the studies selected to women in general, however, appears to be inappropriate. As noted by the authors, prior research published on the prevalence of infertility in other, more general populations of women show much lower values of prevalence than what the authors found in their meta-analysis. This clearly could be related to their decision to limit their analysis to publications that examined both mental health and infertility. This likely included samples from referred populations that hence would not reflect the prevalence of infertility in the population of women in general. Thus, the statement by the authors that the prevalence of infertility that they observed pertains to women in general should be amended to state that this prevalence pertains only to women who participated in studies of mental health and is not consistent with what has been observed in other studies in non-referred, general populations. Also, the risk factors identified in this study are not new, though calling attention to the psychological impact of infertility is worthwhile.

REVIEWER	Valerie Peddie University of Aberdeen, School of Medicine, Dentistry & Nutrition
REVIEW RETURNED	01-Dec-2021

GENERAL COMMENTS	Whilst I am not familiar with the methodology used (statistical software), in my opinion, the authors missed an opportunity when excluding qualitative papers which, in my opinion are best placed to address their 'question'. However, overall, this manuscript is well

	written and if they can justify the exclusion criteria, in addition to addressing why they omitted papers from the EU/UK /US, I would support publication.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

This manuscript presents the results of a random effects model-based meta-analysis of international studies in the literature that examined both psychological distress and infertility. The methodology for the analysis of the data is for the most part consistent with accepted practice, though the I2 value to justify the use of the random effects model was not included in the text and the funnel plots mentioned in the Methods were not shown.

Response from authors: The I squared values were reported in the figures. The statement about funnel plot asymmetry was added for the prevalence of infertility (n = 24).

The generalization of the results based on the studies selected to women in general, however, appears to be inappropriate. As noted by the authors, prior research published on the prevalence of infertility in other, more general populations of women show much lower values of prevalence than what the authors found in their meta-analysis. This clearly could be related to their decision to limit their analysis to publications that examined both mental health and infertility. This likely included samples from referred populations that hence would not reflect the prevalence of infertility in the population of women in general. Thus, the statement by the authors that the prevalence of infertility that they observed pertains to women in general should be amended to state that this prevalence pertains only to women who participated in studies of mental health and is not consistent with what has been observed in other studies in non-referred, general populations.

Response from authors: Thank you. The studies included in the review were focusing on women with infertility. The generalization of the results was specifically for women with that condition. We have maintained the population as females with infertility throughout the text.

Also, the risk factors identified in this study are not new, though calling attention to the psychological impact of infertility is worthwhile.

Response from authors: Yes, we agree that the risk factors were not new. The added value in this review was that we have quantified the strength of statistical associations between infertility and its risk factors.

Reviewer: 2

Whilst I am not familiar with the methodology used (statistical software), in my opinion, the authors missed an opportunity when excluding qualitative papers which, in my opinion are best placed to address their 'question'.

Response from authors: Thank you for your response. This review is intended to determine the prevalence, risk factors, and psychological impact of infertility. The design applied is a systematic

review and meta-analysis, which is applicable for observational quantitative studies. A qualitative study would not fit into this design of meta-analysis. A meta-synthesis would suit better for a review of qualitative studies.

This manuscript is well written and if they can justify the exclusion criteria, in addition to addressing why they omitted papers from the EU/UK /US, I would support publication.
Response from authors: We do not omit papers from the EU/UK /US. We included all full-text and papers published in the English language.

VERSION 2 – REVIEW

REVIEWER	Howard Cabral Boston University School of Public Health, Department of Biostatistics
REVIEW RETURNED	07-Feb-2022

GENERAL COMMENTS	The authors have provided good responses to the first review. Further comments are included below. Original review: This manuscript presents the results of a random effects model-based meta-analysis of international studies in the literature that examined both psychological distress and infertility. The methodology for the analysis of the data is for the most part consistent with accepted practice, though the I2 values to justify the use of the random effects model was not included in the text and the funnel plots mentioned in the Methods were not shown. Response from authors: The I squared values were reported in the figures. The statement about funnel plot asymmetry was added for the prevalence of infertility (n = 24). Further review: Please report the I2 values in the text also. This would be helpful. Were the results of the funnel plots added somewhere in the text, i.e., on page 9 of 50? If so, add text that states,“the results of the assessment of bias were based on the examination of funnel plots (not shown; available upon request)” or something to this effect. Original review: The generalization of the results based on the studies selected to women in general, however, appears to be inappropriate. As noted by the authors, prior research published on the prevalence of infertility in other, more general populations of women show much lower values of prevalence than what the authors found in their meta-analysis. This clearly could be related to their decision to limit their analysis to publications that examined both mental health and infertility. This likely included samples from referred populations that hence would not reflect the prevalence of infertility in the population of women in general. Thus, the statement by the authors that the prevalence of infertility that they observed pertains to women in general should be amended to state that this prevalence pertains only to women who participated in studies of mental health and is not consistent with what has been observed in other studies in non-referred, general populations. Response from authors: Thank you. The studies included in the review were focusing on women with infertility. The generalization of the results was specifically for women with that condition. We have maintained the population as females with infertility throughout the

	text. Further review: For this paper, the authors need to make reference to studies from prior research, how their results are different, and why this may be the case. I assume that the first paragraph of the Discussion section attempts to do this. Is this correct? Original review: Also, the risk factors identified in this study are not new, though calling attention to the psychological impact of infertility is worthwhile. Response from authors: Yes, we agree that the risk factors were not new. The added value in this review was that we have quantified the strength of statistical associations between infertility and its risk factors. Further review: Yes, but the source studies quantified this also and your study is a summary of those estimates, with the caveats that are noted under Limitations.

VERSION 2 – AUTHOR RESPONSE

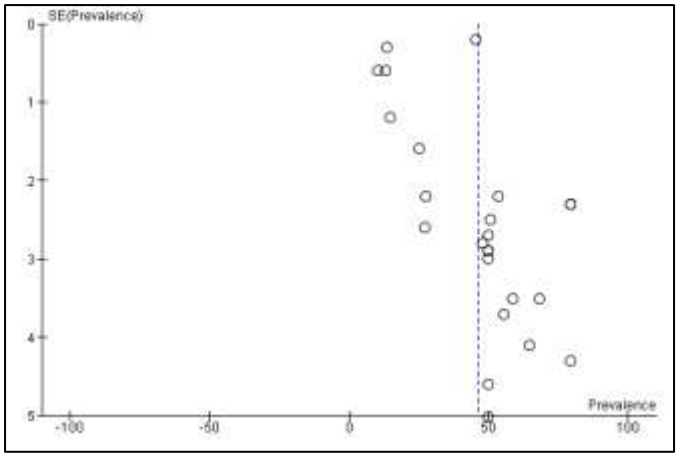
Reviewer: 1
Dr. Howard Cabral, Boston University School of Public Health, Boston University School of Public Health

Comments to the Author:
The authors have provided good responses to the first review. Further comments are included below.

Original review: This manuscript presents the results of a random effects model-based meta-analysis of international studies in the literature that examined both psychological distress and infertility. The methodology for the analysis of the data is for the most part consistent with accepted practice, though the I2 values to justify the use of the random effects model was not included in the text and the funnel plots mentioned in the Methods were not shown.
Response from authors: The I squared values were reported in the figures. The statement about funnel plot asymmetry was added for the prevalence of infertility (n = 24).
Further review: Please report the I2 values in the text also. This would be helpful. Were the results of the funnel plots added somewhere in the text, i.e., on page 9 of 50? If so, add text that states,"the results of the assessment of bias were based on the examination of funnel plots (not shown; available upon request)" or something to this effect.

Response from authors: Thank you. We have added the I^2 values to all the results. The funnel plot result was reported in the Result section under the subheading prevalence as below.

The funnel plot was asymmetry with smaller studies and lower prevalence being missing on the left side. The results of the assessment of bias based on the funnel plot asymmetry were not shown but available on request.



Original review: The generalization of the results based on the studies selected to women in general, however, appears to be inappropriate. As noted by the authors, prior research published on the prevalence of infertility in other, more general populations of women show much lower values of prevalence than what the authors found in their meta-analysis. This clearly could be related to their decision to limit their analysis to publications that examined both mental health and infertility. This likely included samples from referred populations that hence would not reflect the prevalence of infertility in the population of women in general. Thus, the statement by the authors that the prevalence of infertility that they observed pertains to women in general should be amended to state that this prevalence pertains only to women who participated in studies of mental health and is not consistent with what has been observed in other studies in non-referred, general populations.
Response from authors: Thank you. The studies included in the review were focusing on women with infertility. The generalization of the results was specifically for women with that condition. We have maintained the population as females with infertility throughout the text.
Further review: For this paper, the authors need to make reference to studies from prior research, how their results are different, and why this may be the case. I assume that the first paragraph of the Discussion section attempts to do this. Is this correct?

Response from authors: Yes, it is correct. The first paragraph of the Discussion section has cited the previous research and the reason for the differences.

Original review: Also, the risk factors identified in this study are not new, though calling attention to the psychological impact of infertility is worthwhile.
Response from authors: Yes, we agree that the risk factors were not new. The added value in this review was that we have quantified the strength of statistical associations between infertility and its risk factors.
Further review: Yes, but the source studies quantified this also and your study is a summary of those estimates, with the caveats that are noted under Limitations.

Response from authors: Thank you for the suggestion. We have added the following text under Limitations.

Although the risk factors identified in this review are not new, calling attention to the psychological impact of infertility is worthwhile.